Assignment 1) Write a program that demonstrate program structure of java with use of arithmetical and logical implementation.

```
public class Assignment1
{
   public static void main(String[] args)
   {
      // initializing variables
      int num1 = 20, num2 = 10, sum = 0,diff = 0,multi=0;
      float div=0;
      System.out.println("num1 = " + num1);
      System.out.println("num2 = " + num2);
      sum = num1 + num2;
      System.out.println("The sum = " + sum);
      diff = num1 - num2;
      System.out.println("The diff = " + diff);
      multi = num1 * num2;
      System.out.println("The multi = " + multi);
      div = num1 / num2;
      System.out.println("The div = " + div);
      if ((num1==20) && (num2==10))// You can also use || operator
      {
         System.out.println("Both True");
      }
      else
         System.out.println("Both Not True");
   }
   }
```

OUTPUT:-

num1 = 20 num2 = 10 The sum = 30 The diff = 10 The multi = 200 The div = 2.0 Both True

Assignment 2) Write a program that demonstrate string operations using String and StringBuffer class.

```
package assignment2;

import java.io.*;

public class Assignment2

{

   public static void main(String[] args)

   {

      try

            {

        DataInputStream d= new DataInputStream(System.in);

         System.out.println("\n enter the 1st String  ");

         String s=d.readLine();

         //String Functions

         int y=s.length();

             System.out.println("\n length of string is "+y);

             String z=s.toUpperCase();

           System.out.println("\n string in upper case "+z);

           String l=s.toLowerCase();

           System.out.println("\n string in lower  case "+l);

           char m=s.charAt(3);

           System.out.println("\n char at 3rd index is  "+m);

           String o=s.replace('a','b');

           System.out.println("\n replaced string is "+o);

           String n=s.substring(2,5);

           System.out.println("\n sub string from 2 to 5 index  is  "+n);

           System.out.println("\n enter the character to find index");

           String s2=d.readLine();

        int a=s.indexOf(s2);

           System.out.println("\n index of char is "+a);
```

```
System.out.println("\n enter the character to find last index");
    String s3=d.readLine();
    int b=s.lastIndexOf(s3);
    System.out.println("\n last index of char is  "+b);
System.out.println("\n enter the 2nd String ");
String s1=d.readLine();
    String p=s.concat(s1);
    System.out.println("\n concated string is "+p);
    boolean b1=s.equals(s1);
    if(b1==true)
    {
    System.out.println("\n strings are equal ");
    }
    else
    {
      System.out.println("\n strings are  not equal ");
    }
//StringBuffer  Functions
StringBuffer sf = new StringBuffer("Coding Atharva");
System.out.println("\n String = "+sf); // Will Print the string
System.out.println("\n Length = "+sf.length() ); // total numbers of characters
System.out.println("\n Length = "+sf.capacity() ); // total allocated capacity
sf.setLength(6); // Sets the length and destroy the remaining characters
System.out.println("\n After setting length String = "+sf);
sf.setCharAt(0,'K'); // It will change character at specified position
System.out.println("\n SetCharAt String = "+sf);
sf.setCharAt(0,'C');
int a1 = 7;
   sf.append(a1); // It concatenates the other data type value
System.out.println("\n Appended String = "+sf);
```

```
        sf.insert(6," Atharva"); // used to insert one string or char or object
       System.out.println("\n Inserted String = "+sf);
        sf.reverse();
       System.out.println("\n Reverse String = "+sf);
      }
          catch(Exception e)
            {
                    System.out.println(""+e);
            }
     }
   }
```

OUTPUT:-

```
enter the 1st String
manojkumar
 length of string is 10
 string in upper case MANOJKUMAR
 string in lower  case manojkumar
 char at 3rd index is  o
 replaced string is mbnojkumbr
 sub string from 2 to 5 index  is  noj
 enter the character to find index
a
 index of char is 1
 enter the character to find last index
a
 last index of char is  8
 enter the 2nd String
sonawane
 concated string is manojkumarsonawane
 strings are  not equal
```

String = Coding Atharva

Length = 14

Length = 30

After setting length String = Coding

SetCharAt String = Koding

Appended String = Coding7

Inserted String = Coding Atharva7

Reverse String = 7avrahtA gnidoC

Assignment 3) Write a program that demonstrate inner class and static fields.

```
package assignment3;
class Outer
{
   int outer_x = 100;
   void test()
   {
      Inner inner = new Inner(); inner.display();
   }
   static int count=0;//will get memory only once and retain its value
   Outer()
   {
      count++;//incrementing the value of static variable
      System.out.println(count);
   }
  class Inner
{
   void display()
   {
      System.out.println("display: outer_x = " + outer_x);
   }
}
}
public class Assignment3
{
   public static void main(String[] args)
   {
      Outer outer = new Outer();
      outer.test();
      //creating objects
```

```
        Outer o1=new Outer();
        Outer o2=new Outer();
        Outer o3=new Outer();
    }
 }
```

OUTPUT:-

```
1
display: outer_x = 100
2
3
4
```

Assignment 4) Write a program that demonstrate inheritance, polymorphism.

```
package assignment4;
class Animal
{
    public void move()
    {
        System.out.println("Animals can move");
    }
}
class Dog extends Animal
{
    //Method Overriding
    public void move()
    {
        System.out.println("Dogs can walk and run");
    }
    //Method Overloading
    void add(int a,int b)
    {
        int s=a+b;
        System.out.println("Sum="+s);
    }
    void add(int a,int b,int c)
    {
        int s=a+b+c;
        System.out.println("Sum="+s);
    }
 }
public class Assignment4
{
```

```
    public static void main(String[] args)
    {
        Animal a =new Animal();
        Animal b =new Dog();
        a.move();
        b.move();

        Dog d=new Dog();
        d.add(10,20);
        d.add(10,20,30);
    }
}
```

OUTPUT:-

Animals can move

Dogs can walk and run

Sum=30

Sum=60

Assignment 5) Write a program that demonstrate 2D shapes on frames.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag JButtons on JPanel
4. Right Click on JButtons-Edit Text
5. Right Click on JButtons-Events-select event/methods you want and write appropriate code.
6. Code

```
package assignment5;
import java.awt.*;
import java.awt.geom.*;
public class NewJFrame extends javax.swing.JFrame {
    public NewJFrame() {
        initComponents();
    }
     @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jButton4 = new javax.swing.JButton();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        jButton1.setText("Rectangle");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        jButton2.setText("Ellipse");
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
        });

        jButton4.setText("Line");
        jButton4.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton4ActionPerformed(evt);
            }
        });
```

```
        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jButton1)
                .addGap(18, 18, 18)
                .addComponent(jButton2)
                .addGap(18, 18, 18)
                .addComponent(jButton4)
                .addContainerGap(92, Short.MAX_VALUE))
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(27, 27, 27)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
                    .addComponent(jButton1)
                    .addComponent(jButton2)
                    .addComponent(jButton4))
                .addContainerGap(228, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(59, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
```

```
                .addContainerGap())
        );

        pack();
    }// </editor-fold>
    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        Graphics g1=jPanel1.getGraphics();
        Graphics2D g2  = (Graphics2D)g1;
            g2.setPaint(Color.ORANGE);
        double leftx=100;
        double topy=100;
        double width=100;
            double height=200;//For Squre width and height should be same
        Rectangle2D rect = new
Rectangle2D.Double(leftx,topy,leftx+width,topy+height);
            g2.fill(rect);

    }
    private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
         Graphics g1=jPanel1.getGraphics();
        Graphics2D g2  = (Graphics2D)g1;
            g2.setPaint(Color.CYAN);
        double leftx=300;
        double topy=100;
        double width=30;
            double height=40;//For Circle width and height should be same
        Ellipse2D ellipse = new Ellipse2D.Double(leftx,topy,width,height);
            g2.fill(ellipse);
    }
    private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
         Graphics g1=jPanel1.getGraphics();
        Graphics2D g2  = (Graphics2D)g1;
            g2.setPaint(Color.MAGENTA);
        double startx=50;
        double starty=60;
        double endx=600;
            double endy=600;
        Line2D line = new Line2D.Double(startx,starty,endx,endy);
            g2.draw(line);
    }
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
```

```
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (IllegalAccessException ex) {

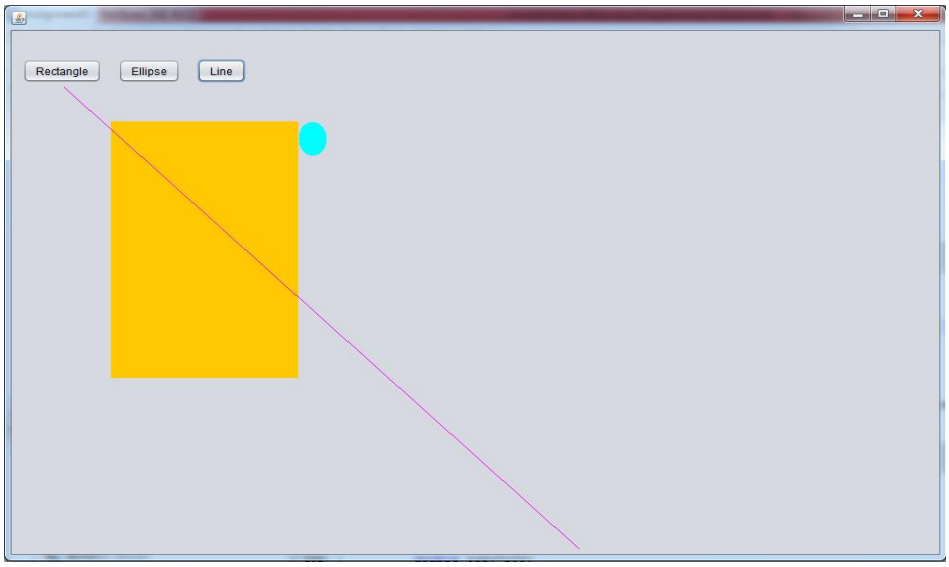
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new NewJFrame().setVisible(true);
            }
        });
    }
    // Variables declaration - do not modify
    private javax.swing.JButton jButton1;
    private javax.swing.JButton jButton2;
    private javax.swing.JButton jButton4;
    private javax.swing.JPanel jPanel1;
    // End of variables declaration
}
```

7. Right Click in Code-Run File

OUTPUT:-



Assignment 6) Write a program that demonstrate color and fonts.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag JButton on JPanel
4. Right Click on JButton-Edit Text
5. Right Click on JButton-Events-select event/methods you want and write appropriate code.
6. Code

```
import java.awt.*;
import java.awt.geom.*;
import java.util.*;
public class NewJFrame extends javax.swing.JFrame {
    public NewJFrame() {
        initComponents();
       }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        jButton1 = new javax.swing.JButton();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        jButton1.setText("Click");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(213, 213, 213)
                .addComponent(jButton1)
                .addContainerGap(667, Short.MAX_VALUE))
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
```

```
            .addComponent(jButton1)
            .addGap(0, 578, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(20, 20, 20)
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(55, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(22, 22, 22)
                .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addContainerGap())
        );

        pack();
    }// </editor-fold>

    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        GraphicsEnvironment
ge=GraphicsEnvironment.getLocalGraphicsEnvironment();
            String s[]=ge.getAvailableFontFamilyNames();
        Graphics g1=jPanel1.getGraphics();
        Random rd = new Random();

        int y=50;
        int sz=20;
                    for(int i=0;i<s.length;i++)
                    {
                            Font f=new Font(s[i],Font.BOLD,sz);//Font.ITALIC
                            g1.setFont(f);

                    int r=rd.nextInt(255);
                    int g=rd.nextInt(255);
                    int b=rd.nextInt(255);
                    Color c=new Color(r,g,b);
                    g1.setColor(c);
```

```
                    g1.drawString("Hello World",50,y);
                y=y+20;
                sz=sz+1;
            }
    }
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

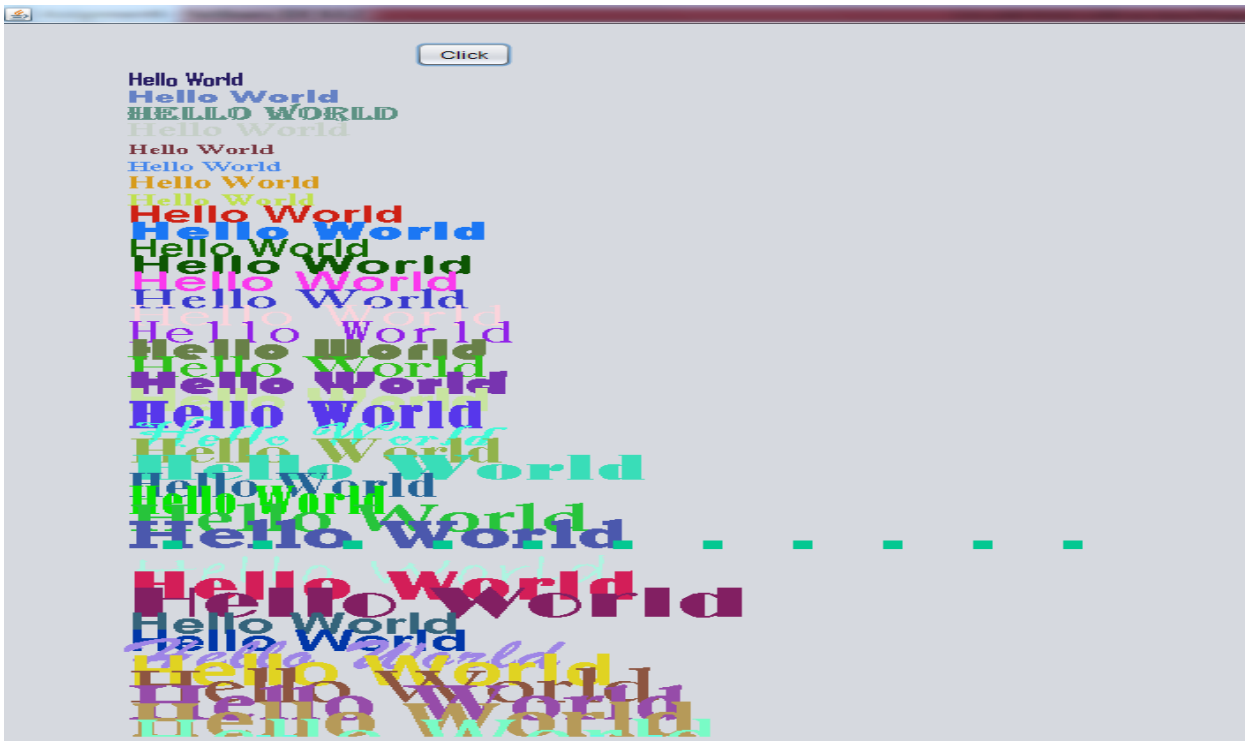
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new NewJFrame().setVisible(true);
            }
        });
```

```
    }
    // Variables declaration - do not modify
    private javax.swing.JButton jButton1;
    private javax.swing.JPanel jPanel1;
    // End of variables declaration
}
```

7. Right Click in Code-Run File

OUTPUT:-


Click
Hello World
Hello World
HELLO WORLD
Hello World
Hello World
Hello World

Assignment 7) Write a program to illustrate use of various swing components.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag various components
4. Right Click on components-Edit Text
5. Drag **ButtonGroup** component and set **buttonGroup** property of radiobuttons**.**
6. Right Click on jComboBox, jList1 and set **model** property.
7. Write Code on Button ActionPerformed

```
package assignment7;
public class NewJFrame extends javax.swing.JFrame {
    public NewJFrame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        buttonGroup1 = new javax.swing.ButtonGroup();
        jPanel1 = new javax.swing.JPanel();
        jLabel1 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jLabel2 = new javax.swing.JLabel();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTextArea1 = new javax.swing.JTextArea();
        jLabel3 = new javax.swing.JLabel();
        jCheckBox1 = new javax.swing.JCheckBox();
        jCheckBox2 = new javax.swing.JCheckBox();
        jCheckBox3 = new javax.swing.JCheckBox();
        jButton1 = new javax.swing.JButton();
        jLabel4 = new javax.swing.JLabel();
        jRadioButton1 = new javax.swing.JRadioButton();
        jRadioButton2 = new javax.swing.JRadioButton();
        jLabel5 = new javax.swing.JLabel();
        jComboBox1 = new javax.swing.JComboBox();
        jLabel6 = new javax.swing.JLabel();
        jScrollPane2 = new javax.swing.JScrollPane();
        jList1 = new javax.swing.JList();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        jLabel1.setText("Enter Rno");

        jLabel2.setText("Enter Name");

        jTextArea1.setColumns(20);
```

```
        jTextArea1.setRows(5);
        jScrollPane1.setViewportView(jTextArea1);

        jLabel3.setText("Favorite Color");

        jCheckBox1.setText("Red");

        jCheckBox2.setText("Green");

        jCheckBox3.setText("Blue");

        jButton1.setText("Click");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });

        jLabel4.setText("Class");

        buttonGroup1.add(jRadioButton1);
        jRadioButton1.setText("MCA-1");

        buttonGroup1.add(jRadioButton2);
        jRadioButton2.setText("MCA-2");

        jLabel5.setText("Laptop");

        jComboBox1.setModel(new javax.swing.DefaultComboBoxModel(new String[]
{ "HP", "Dell", "Lenovo" }));

        jLabel6.setText("Subject");

        jList1.setModel(new javax.swing.AbstractListModel() {
            String[] strings = { "C", "C++", "Java" };
            public int getSize() { return strings.length; }
            public Object getElementAt(int i) { return strings[i]; }
        });
        jScrollPane2.setViewportView(jList1);

        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(35, 35, 35)
```

```
.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                .addGroup(jPanel1Layout.createSequentialGroup()

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
TRAILING, false)
                        .addComponent(jLabel6,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE, 62, Short.MAX_VALUE)
                        .addComponent(jLabel5,
javax.swing.GroupLayout.Alignment.LEADING,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                    .addGap(44, 44, 44)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                        .addComponent(jComboBox1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jScrollPane2,
javax.swing.GroupLayout.PREFERRED_SIZE, 68,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jButton1,
javax.swing.GroupLayout.PREFERRED_SIZE, 92,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                .addGroup(jPanel1Layout.createSequentialGroup()

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                        .addComponent(jLabel1,
javax.swing.GroupLayout.PREFERRED_SIZE, 68,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jLabel2,
javax.swing.GroupLayout.PREFERRED_SIZE, 68,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jLabel3,
javax.swing.GroupLayout.PREFERRED_SIZE, 96,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jLabel4,
javax.swing.GroupLayout.PREFERRED_SIZE, 50,
javax.swing.GroupLayout.PREFERRED_SIZE))

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                        .addGroup(jPanel1Layout.createSequentialGroup()
```

```
                    .addGap(25, 25, 25)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                        .addComponent(jCheckBox1)
                        .addComponent(jCheckBox2)
                        .addComponent(jCheckBox3)
                        .addComponent(jScrollPane1,
javax.swing.GroupLayout.PREFERRED_SIZE, 146,
javax.swing.GroupLayout.PREFERRED_SIZE)
                        .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 89,
javax.swing.GroupLayout.PREFERRED_SIZE)))
                .addGroup(jPanel1Layout.createSequentialGroup()
                    .addGap(13, 13, 13)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                        .addComponent(jRadioButton1)
                        .addComponent(jRadioButton2))))))
            .addContainerGap(691, Short.MAX_VALUE))
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(55, 55, 55)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
                    .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE,
29, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 29,
javax.swing.GroupLayout.PREFERRED_SIZE))

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                    .addComponent(jScrollPane1,
javax.swing.GroupLayout.PREFERRED_SIZE, 62,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE,
25, javax.swing.GroupLayout.PREFERRED_SIZE))

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
```

```
                .addGroup(jPanel1Layout.createSequentialGroup()
                    .addGap(12, 12, 12)
                    .addComponent(jLabel3,
javax.swing.GroupLayout.PREFERRED_SIZE, 35,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addGroup(jPanel1Layout.createSequentialGroup()
                    .addGap(18, 18, 18)
                    .addComponent(jCheckBox1)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                    .addComponent(jCheckBox2)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                    .addComponent(jCheckBox3)))

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                .addGroup(jPanel1Layout.createSequentialGroup()
                    .addGap(21, 21, 21)
                    .addComponent(jRadioButton1)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                    .addComponent(jRadioButton2)
                    .addGap(23, 23, 23))
                .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
jPanel1Layout.createSequentialGroup()

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                    .addComponent(jLabel4,
javax.swing.GroupLayout.PREFERRED_SIZE, 25,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addGap(36, 36, 36)))

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
TRAILING)
                .addComponent(jLabel5, javax.swing.GroupLayout.PREFERRED_SIZE,
26, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(jComboBox1,
javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(18, 18, 18)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                .addComponent(jLabel6, javax.swing.GroupLayout.PREFERRED_SIZE,
24, javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
                .addComponent(jScrollPane2,
javax.swing.GroupLayout.PREFERRED_SIZE, 75,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addGap(31, 31, 31)
            .addComponent(jButton1, javax.swing.GroupLayout.PREFERRED_SIZE,
37, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addContainerGap(64, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(0, 0, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING,
layout.createSequentialGroup()
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE)
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap())
        );

        pack();
    }// </editor-fold>
    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        System.out.println("Rno= "+jTextField1.getText());
        System.out.println("Name= "+jTextArea1.getText());
        String color=" ";
        if (jCheckBox1.isSelected())
        color=color+" "+jCheckBox1.getText();
        if (jCheckBox2.isSelected())
        color=color+" "+jCheckBox2.getText();
        if (jCheckBox3.isSelected())
        color=color+" "+jCheckBox3.getText();

        System.out.println("Favorite Colors= "+color);
```

```
        String cl=" ";
        if (jRadioButton1.isSelected())
        cl=cl+" "+jRadioButton1.getText();
        else
        cl=cl+" "+jRadioButton2.getText();

        System.out.println("Class= "+cl);

        System.out.println("Laptop= "+jComboBox1.getSelectedItem().toString());

        System.out.println("Subjects= ");
        Object o[]=jList1.getSelectedValues();
        for(int i=0;i<o.length;i++)
        {
           System.out.println(o[i].toString());
        }

    }
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (InstantiationException ex) {

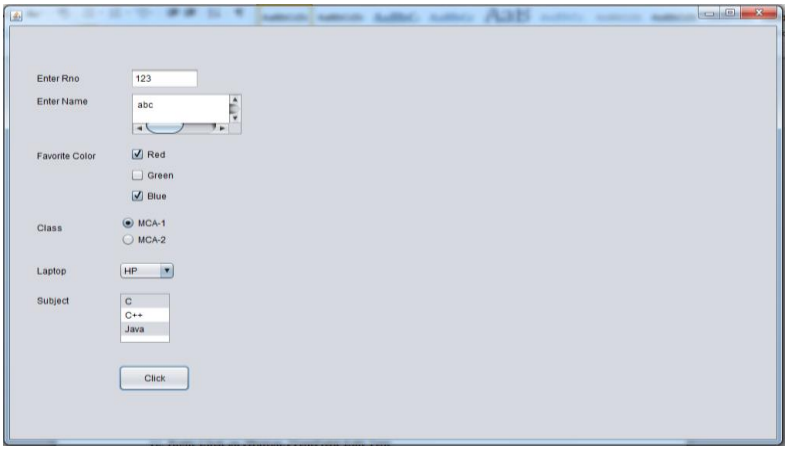
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {
```

```
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        }
        //</editor-fold>
        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new NewJFrame().setVisible(true);
            }
        });
    }
    // Variables declaration - do not modify
    private javax.swing.ButtonGroup buttonGroup1;
    private javax.swing.JButton jButton1;
    private javax.swing.JCheckBox jCheckBox1;
    private javax.swing.JCheckBox jCheckBox2;
    private javax.swing.JCheckBox jCheckBox3;
    private javax.swing.JComboBox jComboBox1;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel jLabel3;
    private javax.swing.JLabel jLabel4;
    private javax.swing.JLabel jLabel5;
    private javax.swing.JLabel jLabel6;
    private javax.swing.JList jList1;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JRadioButton jRadioButton1;
    private javax.swing.JRadioButton jRadioButton2;
    private javax.swing.JScrollPane jScrollPane1;
    private javax.swing.JScrollPane jScrollPane2;
    private javax.swing.JTextArea jTextArea1;
    private javax.swing.JTextField jTextField1;
    // End of variables declaration
}
```

8. Right Click in Code-Run File

OUTPUT:-



Assignment 8) Write a program that demonstrate use of dialog box and menus.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag **JMenuBar**--Edit Text
4. Right Click on JMenuBar-select **Add From Palette-MenuItem/Separator**.
5. Right Click on MenuItem-select event/methods you want.
6. Drag **Popup Menu** on JPanel and add MenuItem, event/methods in it similarly.
7. Right Click on your JPanel, set **componentPopupMenu** property to your popup menu.
8. For User DialogBox- Drag **JDialog** on JPanel, Right Click on your JDialog-**setLayout,** Right Click on your JDialog-**Add From Palette-Swing Controls.**
9. Write Following Code

```
package assignment8;
import javax.swing.*;
import java.io.*;
import java.awt.*;
public class NewJFrame extends javax.swing.JFrame {
   public NewJFrame() {
      initComponents();
   }
   @SuppressWarnings("unchecked")
   // <editor-fold defaultstate="collapsed" desc="Generated Code">
   private void initComponents() {

      jPopupMenu1 = new javax.swing.JPopupMenu();
      Red = new javax.swing.JMenuItem();
      Green = new javax.swing.JMenuItem();
      Blue = new javax.swing.JMenuItem();
      jDialog1 = new javax.swing.JDialog();
      jTextField1 = new javax.swing.JTextField();
      Click = new javax.swing.JButton();
      jPanel1 = new javax.swing.JPanel();
      jMenuBar1 = new javax.swing.JMenuBar();
      jMenu1 = new javax.swing.JMenu();
      jMenuItem1 = new javax.swing.JMenuItem();
      jSeparator1 = new javax.swing.JPopupMenu.Separator();
      jMenuItem2 = new javax.swing.JMenuItem();
      jSeparator2 = new javax.swing.JPopupMenu.Separator();
      jCheckBoxMenuItem1 = new javax.swing.JCheckBoxMenuItem();
      jSeparator3 = new javax.swing.JPopupMenu.Separator();
      jRadioButtonMenuItem1 = new javax.swing.JRadioButtonMenuItem();
      jSeparator5 = new javax.swing.JPopupMenu.Separator();
      jMenuItem6 = new javax.swing.JMenuItem();
      jSeparator4 = new javax.swing.JPopupMenu.Separator();
```

```
        jMenuItem4 = new javax.swing.JMenuItem();
        jMenu2 = new javax.swing.JMenu();
        jMenuItem3 = new javax.swing.JMenuItem();
        Red.setText("Red");
        Red.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                RedActionPerformed(evt);
            }
        });
        jPopupMenu1.add(Red);

        Green.setText("Green");
        Green.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                GreenActionPerformed(evt);
            }
        });
        jPopupMenu1.add(Green);
        Blue.setText("Blue");
        Blue.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                BlueActionPerformed(evt);
            }
        });
        jPopupMenu1.add(Blue);
        jDialog1.getContentPane().setLayout(new java.awt.FlowLayout());
        jTextField1.setText("jTextField1");
        jDialog1.getContentPane().add(jTextField1);
        Click.setText("Click");
        jDialog1.getContentPane().add(Click);
        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);
        jPanel1.setComponentPopupMenu(jPopupMenu1);
        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGap(0, 958, Short.MAX_VALUE)
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGap(0, 581, Short.MAX_VALUE)
        );

        jMenu1.setText("File");
        jMenu1.addActionListener(new java.awt.event.ActionListener() {
```

```
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenu1ActionPerformed(evt);
            }
        });


jMenuItem1.setAccelerator(javax.swing.KeyStroke.getKeyStroke(java.awt.event.Key
Event.VK_A, java.awt.event.InputEvent.CTRL_MASK));
        jMenuItem1.setText("InputDialogBox");
        jMenuItem1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem1ActionPerformed(evt);
            }
        });
        jMenu1.add(jMenuItem1);
        jMenu1.add(jSeparator1);


jMenuItem2.setAccelerator(javax.swing.KeyStroke.getKeyStroke(java.awt.event.Key
Event.VK_B, java.awt.event.InputEvent.CTRL_MASK));
        jMenuItem2.setText("MessageDialogBox");
        jMenuItem2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem2ActionPerformed(evt);
            }
        });
        jMenu1.add(jMenuItem2);
        jMenu1.add(jSeparator2);


jCheckBoxMenuItem1.setAccelerator(javax.swing.KeyStroke.getKeyStroke(java.awt.
event.KeyEvent.VK_C, java.awt.event.InputEvent.ALT_MASK));
        jCheckBoxMenuItem1.setSelected(true);
        jCheckBoxMenuItem1.setText("ConfirmDialogBox");
        jCheckBoxMenuItem1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jCheckBoxMenuItem1ActionPerformed(evt);
            }
        });
        jMenu1.add(jCheckBoxMenuItem1);
        jMenu1.add(jSeparator3);


jRadioButtonMenuItem1.setAccelerator(javax.swing.KeyStroke.getKeyStroke(java.a
wt.event.KeyEvent.VK_D, java.awt.event.InputEvent.SHIFT_MASK));
        jRadioButtonMenuItem1.setSelected(true);
        jRadioButtonMenuItem1.setText("OptionDialogBox");
```

```
        jRadioButtonMenuItem1.addActionListener(new
java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jRadioButtonMenuItem1ActionPerformed(evt);
            }
        });
        jMenu1.add(jRadioButtonMenuItem1);
        jMenu1.add(jSeparator5);

        jMenuItem6.setText("FileChooser");
        jMenuItem6.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem6ActionPerformed(evt);
            }
        });
        jMenu1.add(jMenuItem6);
        jMenu1.add(jSeparator4);

        jMenuItem4.setText("ColorChooser");
        jMenuItem4.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem4ActionPerformed(evt);
            }
        });
        jMenu1.add(jMenuItem4);
        jMenuBar1.add(jMenu1);
        jMenu2.setText("Edit");
        jMenuItem3.setText("UserDialogBox");
        jMenuItem3.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jMenuItem3ActionPerformed(evt);
            }
        });
        jMenu2.add(jMenuItem3);
        jMenuBar1.add(jMenu2);

        setJMenuBar(jMenuBar1);
        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(28, 28, 28)
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(26, Short.MAX_VALUE))
```

```
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(35, 35, 35)
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE,
Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>
    private void jMenuItem1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        String n=JOptionPane.showInputDialog("Enter Name");
        System.out.println("Name="+n);
    }
    private void jMenuItem3ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        jDialog1.setTitle("This is my DialogBox");
        jDialog1.setSize(222,222);
        jDialog1.show();
    }
    private void jMenuItem2ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        JOptionPane.showMessageDialog(null,"Success");

    }
    private void jCheckBoxMenuItem1ActionPerformed(java.awt.event.ActionEvent
evt) {
        // TODO add your handling code here:
        int i=JOptionPane.showConfirmDialog(null, "Are you Sure?");
        System.out.println(i);
    }
    private void jRadioButtonMenuItem1ActionPerformed(java.awt.event.ActionEvent
evt) {
        // TODO add your handling code here:
        String[] options = {"first", "second", "third"};
        int x = JOptionPane.showOptionDialog(null, "Select Option",
                "OptionDialogBox",JOptionPane.DEFAULT_OPTION,
JOptionPane.INFORMATION_MESSAGE, null, options, options[0]);
        System.out.println("Your Option is "+x);

    }
    private void RedActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
```

```
        jPanel1.setBackground(Color.red);
    }
    private void GreenActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        jPanel1.setBackground(Color.green);
    }
    private void BlueActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        jPanel1.setBackground(Color.blue);
    }
    private void jMenu1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:

    }
    private void jMenuItem6ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        JFileChooser fc=new JFileChooser();
        int i=fc.showOpenDialog(this);
        if(i==JFileChooser.APPROVE_OPTION)
        {
             File f=fc.getSelectedFile();
            String filepath=f.getPath();
            System.out.println("You Selected "+filepath);
        }
    }
    private void jMenuItem4ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        Color c=JColorChooser.showDialog(this,"Select a color",Color.ORANGE);
        jPanel1.setBackground(c);

    }
    public static void main(String args[]) {
        /* Set the Nimbus look and feel */
        //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
        /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
         * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
         */
        try {
            for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                if ("Nimbus".equals(info.getName())) {
                    javax.swing.UIManager.setLookAndFeel(info.getClassName());
                    break;
                }
            }
```

```
        } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

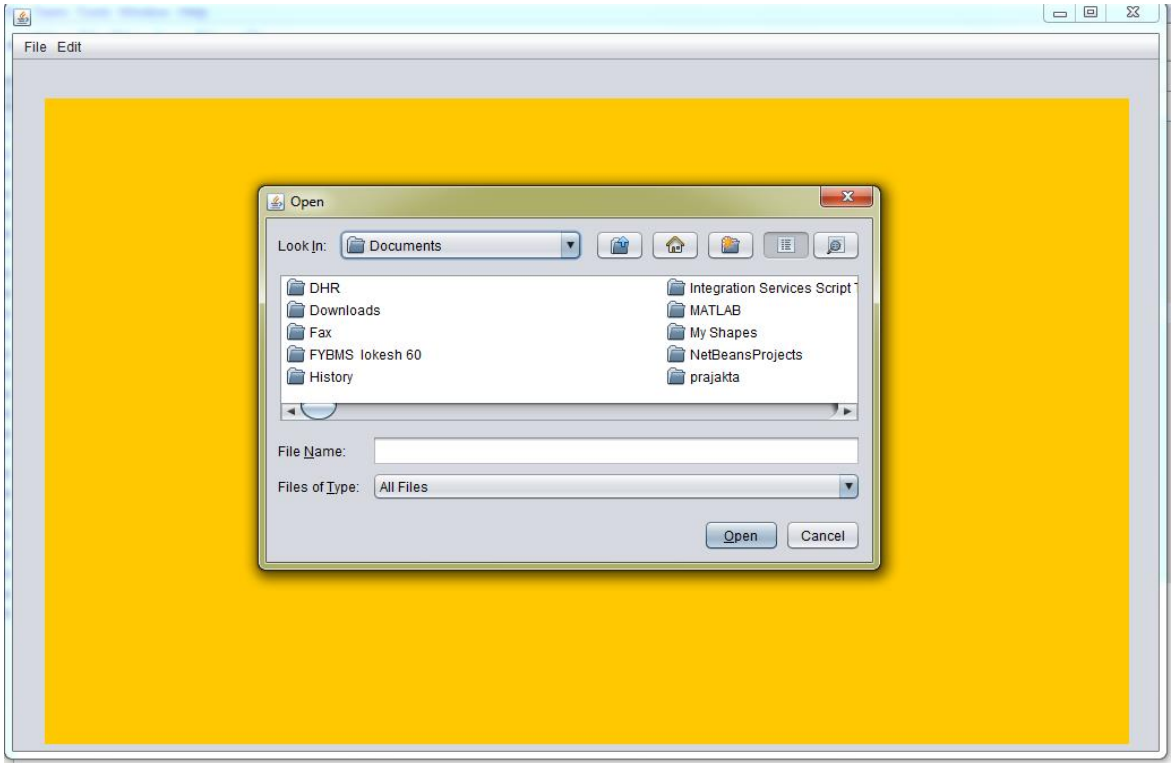
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new NewJFrame().setVisible(true);
            }
        });
    }
    // Variables declaration - do not modify
    private javax.swing.JMenuItem Blue;
    private javax.swing.JButton Click;
    private javax.swing.JMenuItem Green;
    private javax.swing.JMenuItem Red;
    private javax.swing.JCheckBoxMenuItem jCheckBoxMenuItem1;
    private javax.swing.JDialog jDialog1;
    private javax.swing.JMenu jMenu1;
    private javax.swing.JMenu jMenu2;
    private javax.swing.JMenuBar jMenuBar1;
    private javax.swing.JMenuItem jMenuItem1;
    private javax.swing.JMenuItem jMenuItem2;
    private javax.swing.JMenuItem jMenuItem3;
    private javax.swing.JMenuItem jMenuItem4;
    private javax.swing.JMenuItem jMenuItem6;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JPopupMenu jPopupMenu1;
    private javax.swing.JRadioButtonMenuItem jRadioButtonMenuItem1;
    private javax.swing.JPopupMenu.Separator jSeparator1;
    private javax.swing.JPopupMenu.Separator jSeparator2;
    private javax.swing.JPopupMenu.Separator jSeparator3;
    private javax.swing.JPopupMenu.Separator jSeparator4;
```

```
        private javax.swing.JPopupMenu.Separator jSeparator5;
        private javax.swing.JTextField jTextField1;
        // End of variables declaration
    }
```

10. Right Click in Code-Run File

OUTPUT:-



Assignment 9) Write a program that demonstrate event handling for various types of events.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag JButton, JTextField on JPanel
4. Right Click on JButton, JTextField-Edit Text
5. Right Click on JButton, JTextField, JPanel-Events-select event/methods you want and write appropriate code.
6. Code

```
package assignment9;
import java.awt.Color;
public class NewJFrame extends javax.swing.JFrame {
    public NewJFrame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        jButton2 = new javax.swing.JButton();
        jTextField1 = new javax.swing.JTextField();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        jPanel1.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
                jPanel1MouseClicked(evt);
            }
        });

        jButton2.setText("Mouse");
        jButton2.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseEntered(java.awt.event.MouseEvent evt) {
                jButton2MouseEntered(evt);
            }
            public void mouseExited(java.awt.event.MouseEvent evt) {
                jButton2MouseExited(evt);
            }
        });

        jTextField1.addKeyListener(new java.awt.event.KeyAdapter() {
            public void keyTyped(java.awt.event.KeyEvent evt) {
                jTextField1KeyTyped(evt);
            }
        });
```

```
        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(53, 53, 53)
                .addComponent(jButton2, javax.swing.GroupLayout.PREFERRED_SIZE,
112, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addGap(81, 81, 81)
                .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 95,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(635, Short.MAX_VALUE))
        );
        jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(24, 24, 24)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
                    .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 33,
javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 33,
javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(541, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(19, Short.MAX_VALUE))
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
```

```
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(28, Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>

    private void jButton2MouseEntered(java.awt.event.MouseEvent evt) {
        // TODO add your handling code here:
        jPanel1.setBackground(Color.red);
    }
    private void jButton2MouseExited(java.awt.event.MouseEvent evt) {
        // TODO add your handling code here:
         jPanel1.setBackground(Color.GREEN);
    }
    private void jTextField1KeyTyped(java.awt.event.KeyEvent evt) {
        // TODO add your handling code here:
         char a=evt.getKeyChar();
                        if(a=='r' || a=='R')
                        {
                                jPanel1.setBackground(Color.red);

                        }
                else if(a=='g' || a=='G')
                        {
                          jPanel1.setBackground(Color.GREEN);

                        }
                else
                {
                      jPanel1.setBackground(Color.BLACK);
                }
    }
 int count=0;
    private void jPanel1MouseClicked(java.awt.event.MouseEvent evt) {
        // TODO add your handling code here:
        count++;
        if(count==1)
           jPanel1.setBackground(Color.RED);
        else if(count==2)
           jPanel1.setBackground(Color.GREEN);
        else if(count==3)
           jPanel1.setBackground(Color.BLUE);
        else
```

```
            count=0;
        }
        public static void main(String args[]) {
            /* Set the Nimbus look and feel */
            //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
            /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
             * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
             */
            try {
                for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
                    if ("Nimbus".equals(info.getName())) {
                        javax.swing.UIManager.setLookAndFeel(info.getClassName());
                        break;
                    }
                }
            } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
            } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
            } catch (IllegalAccessException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
            } catch (javax.swing.UnsupportedLookAndFeelException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
            }
            //</editor-fold>

            /* Create and display the form */
            java.awt.EventQueue.invokeLater(new Runnable() {
                public void run() {
                    new NewJFrame().setVisible(true);
                }
            });
        }

        // Variables declaration - do not modify
        private javax.swing.JButton jButton2;
```

```
    private javax.swing.JPanel jPanel1;
    private javax.swing.JTextField jTextField1;
    // End of variables declaration
}
```

7. Right Click in Code-Run File

OUTPUT:-



Assignment 10) Write a program to illustrate multithreading.

```
package assignment10;
class TestSleepMethod1 extends Thread
{
   public void run()
   {
      for(int i=1;i<=5;i++)
      {
         try
         {
            System.out.println(i);
            Thread.sleep(500);
         }
         catch(InterruptedException e)
         {
            System.out.println(e);
         }
       }
   }
}
public class Assignment10
{
   public static void main(String[] args)
   {
      TestSleepMethod1 t1=new TestSleepMethod1();
      TestSleepMethod1 t2=new TestSleepMethod1();
      TestSleepMethod1 t3=new TestSleepMethod1();
      t1.start();
      t2.start();
      t3.start();
```

```
    }
}
```

OUTPUT:-

```
1
1
1
2
2
2
3
3
3
4
4
4
5
5
5
```

Assignment 11) Write a program to illustrate exception handling.

```
package assignment11;
public class Assignment11
{
   public static void main(String[] args)
   {
      try
      {
         int i=2/0;
         int a[]=new int[5];
         a[10]=30;
      }
      catch(ArrayIndexOutOfBoundsException e)
      {
         System.out.println("ArrayIndexOutOfBoundsException");
      }
      catch(ArithmeticException e)
      {
         System.out.println("ArithmeticException");
      }
      catch(Exception e)
      {
         System.out.println("Exception");
      }
      finally
      {
         System.out.println("Finally");
      }
   }
 }
```

OUTPUT:-

ArithmeticException

Finally

Assignment 12) Write a program to demonstrate use of File class.

```
package assignment12;
import java.io.*;
public class Assignment12
{
   public static void main(String[] args)
   {
      FileInputStream sourceStream = null; //FileReader for char by char
      FileOutputStream targetStream = null; //FileWriter for char by char
      try
      {
         sourceStream= new FileInputStream("sorcefile.txt");
         targetStream= new FileOutputStream("targetfile.txt");
         // Reading source file and writing
         // content to target file byte by byte
         int temp;
         while ((temp = sourceStream.read())!= -1)
         {
            targetStream.write(temp);
         }
         sourceStream.close();
         targetStream.close();

      }
      catch(Exception e)
      {
         System.out.println("Exception");
      }
       //File class
       File f = new File("sorcefile.txt");
```

```
        System.out.println("The name of the file is: " + f.getName());
        System.out.println("The absolute path of the file is: " + f.getAbsolutePath());
        System.out.println("Is file writeable?: " + f.canWrite());
        System.out.println("Is file readable " + f.canRead());
        System.out.println("The size of the file in bytes is: " + f.length());
        System.out.println("File Exist "+f.exists());
        System.out.println("Is File or Directory "+f.isFile());
        System.out.println("Is File or Directory "+f.isDirectory());
        System.out.println("Is Hidden "+f.isHidden());
        System.out.println("Last Modified Time: " + f.lastModified());
      }
    }
```

OUTPUT:

The name of the file is: sorcefile.txt

The absolute path of the file is: F:\M.S.Sonawane\2021-22\Java\Assignment12\sorcefile.txt

Is file writeable?: true

Is file readable true

The size of the file in bytes is: 46

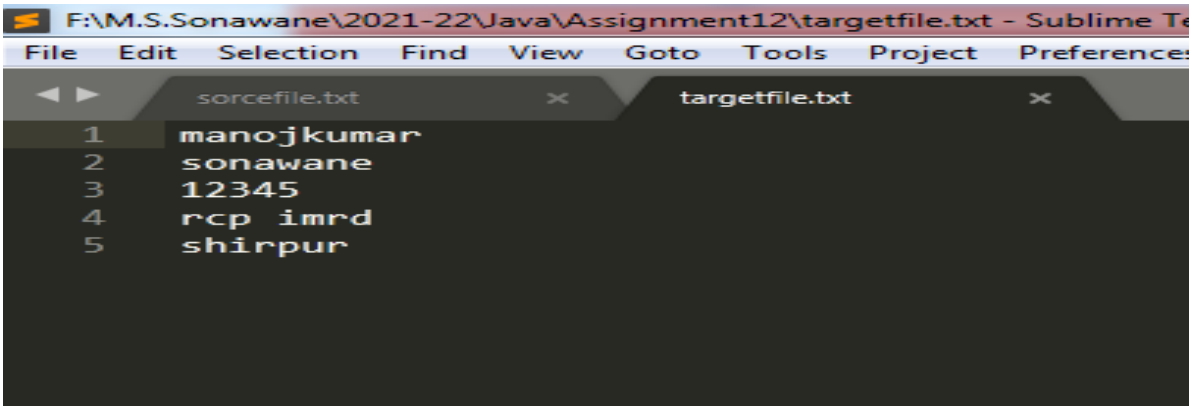
File Exist true

Is File or Directory true

Is File or Directory false

Is Hidden false

Last Modified Time: 1642157554913



Assignment 13) Write a program that demonstrate JDBC on application.

Steps:-

1. Right Click on your project- New-JFrame
2. Drag JPanel on JFrame
3. Drag 2 JLabels, 2 JTextFields, 4 JButtons on JPanel
4. Right Click on all-Edit Text
5. Create Database
6. Create DSN and connect it to Database.
7. Connect DSN to your application in NetBeans.
8. Right Click on 4 JButtons-Events-select event/methods you want and write appropriate code.
9. Code

```
package assignment13;
import java.sql.*;
public class NewJFrame extends javax.swing.JFrame {
    public NewJFrame() {
        initComponents();
    }
    @SuppressWarnings("unchecked")
    // <editor-fold defaultstate="collapsed" desc="Generated Code">
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        jLabel1 = new javax.swing.JLabel();
        jTextField1 = new javax.swing.JTextField();
        jLabel2 = new javax.swing.JLabel();
        jTextField2 = new javax.swing.JTextField();
        jButton1 = new javax.swing.JButton();
        jButton2 = new javax.swing.JButton();
        jButton3 = new javax.swing.JButton();
        jButton4 = new javax.swing.JButton();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        jLabel1.setText("RNo");

        jLabel2.setText("Name");

        jButton1.setText("Insert");
        jButton1.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton1ActionPerformed(evt);
            }
        });
```

```
        jButton2.setText("Update");
        jButton2.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton2ActionPerformed(evt);
            }
        });

        jButton3.setText("Delete");
        jButton3.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton3ActionPerformed(evt);
            }
        });

        jButton4.setText("Select");
        jButton4.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                jButton4ActionPerformed(evt);
            }
        });

        javax.swing.GroupLayout jPanel1Layout = new
javax.swing.GroupLayout(jPanel1);
        jPanel1.setLayout(jPanel1Layout);
        jPanel1Layout.setHorizontalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addGap(83, 83, 83)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING, false)
                    .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE,
72, Short.MAX_VALUE)
                    .addComponent(jLabel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE,
53, javax.swing.GroupLayout.PREFERRED_SIZE))

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING)
                    .addGroup(jPanel1Layout.createSequentialGroup()
                        .addComponent(jButton2,
javax.swing.GroupLayout.PREFERRED_SIZE, 83,
javax.swing.GroupLayout.PREFERRED_SIZE)
```

```
.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                .addComponent(jButton3,
javax.swing.GroupLayout.PREFERRED_SIZE, 81,
javax.swing.GroupLayout.PREFERRED_SIZE)

.addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.UNRELATED)
                .addComponent(jButton4,
javax.swing.GroupLayout.PREFERRED_SIZE, 89,
javax.swing.GroupLayout.PREFERRED_SIZE))
            .addComponent(jTextField2,
javax.swing.GroupLayout.PREFERRED_SIZE, 106,
javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 74,
javax.swing.GroupLayout.PREFERRED_SIZE))
          .addContainerGap(569, Short.MAX_VALUE))
    );
    jPanel1Layout.setVerticalGroup(

jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel1Layout.createSequentialGroup()
          .addGap(56, 56, 56)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
            .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE,
23, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jTextField1,
javax.swing.GroupLayout.PREFERRED_SIZE, 23,
javax.swing.GroupLayout.PREFERRED_SIZE))
          .addGap(33, 33, 33)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
            .addComponent(jLabel2, javax.swing.GroupLayout.PREFERRED_SIZE,
26, javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jTextField2,
javax.swing.GroupLayout.PREFERRED_SIZE, 26,
javax.swing.GroupLayout.PREFERRED_SIZE))
          .addGap(62, 62, 62)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
LEADING, false)

.addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.
BASELINE)
```

```
                    .addComponent(jButton2,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jButton3,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(jButton4,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                .addComponent(jButton1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
            .addContainerGap(362, Short.MAX_VALUE))
        );

        javax.swing.GroupLayout layout = new
javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                .addContainerGap())
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(jPanel1, javax.swing.GroupLayout.PREFERRED_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(42, Short.MAX_VALUE))
        );

        pack();
    }// </editor-fold>

    private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {
        // TODO add your handling code here:
        try
          {
          Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
         Connection c=DriverManager.getConnection("jdbc:odbc:dsn1"," "," ");
          Statement st=c.createStatement();;
          String s1=jTextField1.getText();
          int i=Integer.parseInt(s1);
          String s2=jTextField2.getText();
```

```
        int count=st.executeUpdate("insert into student values("+i+",'"+s2+"')");
        System.out.println("Record Inserted "+count);
        }
      catch(Exception e)
        {
          System.out.println("Insert Exp "+e);
        }
    }
    private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      try
        {
        Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
       Connection c=DriverManager.getConnection("jdbc:odbc:dsn1"," "," ");
        Statement st=c.createStatement();;
        String s1=jTextField1.getText();
        int i=Integer.parseInt(s1);
        String s2=jTextField2.getText();
        int count=st.executeUpdate("update student set sname='"+s2+"' where
rno="+i+"");
        System.out.println("Record Updated "+count);
        }
      catch(Exception e)
        {
          System.out.println("Update Exp "+e);
        }
    }
    private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      try
        {
        Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
       Connection c=DriverManager.getConnection("jdbc:odbc:dsn1"," "," ");
        Statement st=c.createStatement();;
        String s1=jTextField1.getText();
        int i=Integer.parseInt(s1);
        int count=st.executeUpdate("delete * from student where rno="+i+"");
        System.out.println("Record Deleted "+count);
        }
      catch(Exception e)
        {
          System.out.println("Delete Exp "+e);
        }
    }
    private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      try
        {
```

```
        Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
        Connection c=DriverManager.getConnection("jdbc:odbc:dsn1"," "," ");
        Statement st=c.createStatement();;
        String s1=jTextField1.getText();
        int i=Integer.parseInt(s1);
        ResultSet rs=st.executeQuery("select * from student where rno="+i+"");
        while(rs.next())
        {
              jTextField2.setText(rs.getString("sname"));
         }
        }
      catch(Exception e)
         {
            System.out.println("Select Exp "+e);
         }
   }

   /**
    * @param args the command line arguments
    */
   public static void main(String args[]) {
      /* Set the Nimbus look and feel */
      //<editor-fold defaultstate="collapsed" desc=" Look and feel setting code
(optional) ">
      /* If Nimbus (introduced in Java SE 6) is not available, stay with the default look
and feel.
       * For details see
http://download.oracle.com/javase/tutorial/uiswing/lookandfeel/plaf.html
       */
      try {
         for (javax.swing.UIManager.LookAndFeelInfo info :
javax.swing.UIManager.getInstalledLookAndFeels()) {
            if ("Nimbus".equals(info.getName())) {
               javax.swing.UIManager.setLookAndFeel(info.getClassName());
               break;
            }
         }
      } catch (ClassNotFoundException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
      } catch (InstantiationException ex) {

java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
      } catch (IllegalAccessException ex) {
```

```
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        } catch (javax.swing.UnsupportedLookAndFeelException ex) {

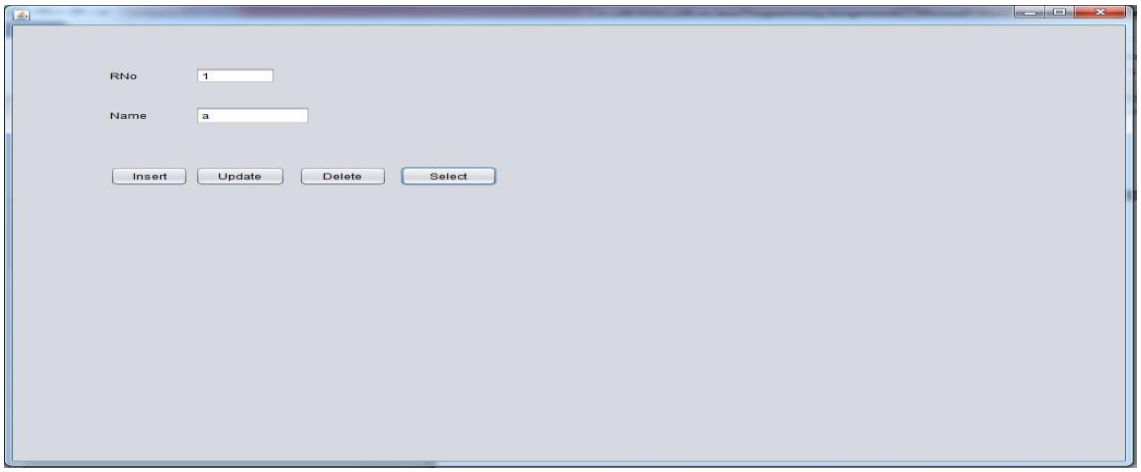
java.util.logging.Logger.getLogger(NewJFrame.class.getName()).log(java.util.loggin
g.Level.SEVERE, null, ex);
        }
        //</editor-fold>

        /* Create and display the form */
        java.awt.EventQueue.invokeLater(new Runnable() {
            public void run() {
                new NewJFrame().setVisible(true);
            }
        });
    }

    // Variables declaration - do not modify
    private javax.swing.JButton jButton1;
    private javax.swing.JButton jButton2;
    private javax.swing.JButton jButton3;
    private javax.swing.JButton jButton4;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JTextField jTextField1;
    private javax.swing.JTextField jTextField2;
    // End of variables declaration
}
```

10. Right Click in Code-Run File

OUTPUT:-



Assignment 14) Write a program that demonstrates package creation and use in program.

```
package assignment14;
import mypackage.NewClass;
public class Assignment14
{
    public static void main(String[] args)
    {
        NewClass n=new NewClass();
        n.show();
    }
  }
//Create mypackage, Create NewClass
package mypackage;
public class NewClass
{
    public void show()
    {
        System.out.println("Show Method is Called");
    }
}
```

OUTPUT:-

```
Show Method is Called
```